Neuberg DIAGNOSTICS | CENTER FOR GENOMIC MEDICINE

ORION Intelligent Genomics

Patient Details

Name	:	DEVIKA	**Sex / Age**	:	Female/ 19 Years	**Case ID**	:	60300129567
Ref By	:		**PT. ID**	:	2908914 OQG2603250447	**Test Name**	:	ORION (V2) (WES-Whole Exome Sequencing)
Bill. Loc.	:	Oncquest Laboratories Ltd.						

Sample Details

Registration Date & Time	:	2026-03-25 16:51:18	**Sample Type**	:	Whole Blood EDTA	**Sample Date & Time**	:	2026-03-25 16:51:00
Ref ID 1.	:	-	**Report Date & Time**	:	2026-04-19 14:45:21			

Summary of the Clinical History Provided

Salient features: Chronic calcific pancreatitis, intrapancreatic ductal calculi, polypoidal lesion in the distal pancreatic duct
Clinical suspicion: Pancreatic exocrine insufficiency

Test Results and Interpretation

ON PHENOTYPE BASED ANALYSIS: VARIANT/S DETECTED.

PRSS1 variants are low-depth, low-complexity variants, and hence confirmation via sanger sequencing is recommended.

Summary Of Variants

Gene and Transcript	Exon/Intron Number	Variant Nomenclature	Zygosity	Classification	OMIM Phenotype	Inheritance
CTRC (NM_007272.3)	Exon 3/Exon 8 -	c.217G>A p.Ala73Thr [101x /186x]	Heterozygous	Pathogenic	{Pancreatitis, chronic, susceptibility to}	Autosomal dominant
PRSS1 (NM_002769.5)	Exon 1/Exon 5 -	c.40C>G p.Leu14Val [54x /243x]	Heterozygous	Uncertain significance	Pancreatitis, hereditary	Autosomal dominant
PRSS1 (NM_002769.5)	Exon 2/Exon 5 -	c.86A>T p.Asn29Ile [85x /531x]	Heterozygous	Pathogenic	Pancreatitis, hereditary	Autosomal dominant

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Variant Details

CTRC

Variant Nomenclature	c.217G>A (p.Ala73Thr)
Genomic Nomenclature	chr1:g.15767073G>A
Zygosity	Heterozygous

Type of variant	gnomAD frequency	Computational evidences	ClinVar	Null Variant	Downstream Null Variant	Previously reported [reported zygosity]
Missense Variant	0.0285%	**REVEL:** 0.972	Yes (Pathogenic/ Likely Pathogenic/ Uncertain Significance)	NA	NA	Yes (heterozygous)

Experimental studies have shown that this missense change affects CTRC function (Rosendahl J, et al., 2008).

References:

1. Berke G,et al. Risk of chronic pancreatitis in carriers of the c.180C>T (p.Gly60=) CTRC variant: case-control studies and metaanalysis. Pancreatology. 2023.
2. Rosendahl J, et al. Chymotrypsin C (CTRC) variants that diminish activity or secretion are associated with chronic pancreatitis. Nat Genet. 2008.

Disease

PANCREATITIS, HEREDITARY [OMIM GENE ID: 601405]

PRSS1-related hereditary pancreatitis (HP) is characterized by episodes of acute pancreatitis (AP) and recurrent acute pancreatitis (RAP: >1 episode of AP), with frequent progression to chronic pancreatitis (CP). Mani from vague abdominal pain lasting one to three days to severe abdominal pain lasting days to weeks and requiring hospitalization.

PRSS1

Variant Nomenclature	c.40C>G (p.Leu14Val)
Genomic Nomenclature	chr7:g.142457375C>G
Zygosity	Heterozygous

Type of variant	gnomAD frequency	Computational evidences	ClinVar	Null Variant	Downstream Null Variant	Previously reported [reported zygosity]
Missense Variant, Splice Region Variant	1.6079%	**REVEL:** 0.221	Yes (Uncertain significance; Benign; Likely benign	NA	NA	Yes (Internal Database)

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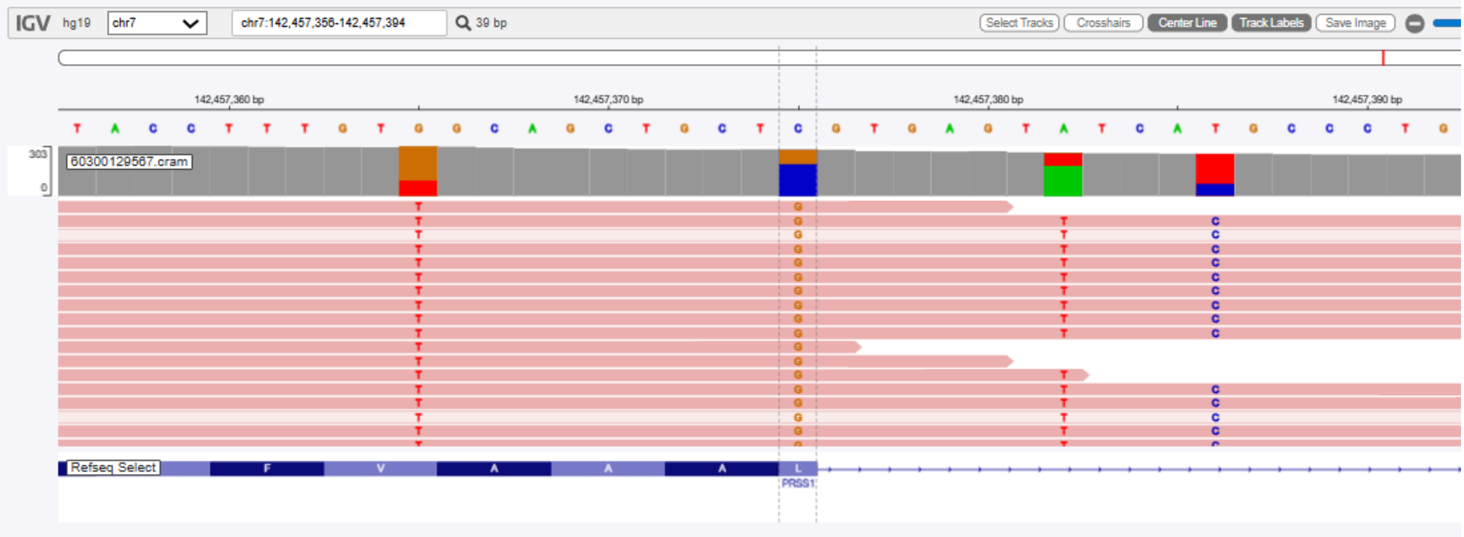
Variant Details



Disease

PANCREATITIS, HEREDITARY [OMIM GENE ID: 276000]

PRSS1-related hereditary pancreatitis (HP) is characterized by episodes of acute pancreatitis (AP) and recurrent acute pancreatitis (RAP: >1 episode of AP), with frequent progression to chronic pancreatitis (CP). Mani from vague abdominal pain lasting one to three days to severe abdominal pain lasting days to weeks and requiring hospitalization.

PRSS1

Variant Nomenclature	c.86A>T (p.Asn29Ile)
Genomic Nomenclature	chr7:g.142458451A>T
Zygosity	Heterozygous

Type of variant	gnomAD frequency	Computational evidences	ClinVar	Null Variant	Downstream Null Variant	Previously reported [reported zygosity]
Missense Variant	0.0003%	**REVEL:** 0.269	Pathogenic/ Uncertain Significance	NA	NA	Yes [Heterozygous]

This variant has been observed to segregate with disease in related individuals (Gorry MC, et al; 1997). Experimental studies have shown that this missense change affects PRSS1 fu M. et al; 2000).

References:

1. Gorry MC, et al; Mutations in the cationic trypsinogen gene are associated with recurrent acute and chronic pancreatitis. Gastroenterology. 1997 Oct;113(4):1063-8.
2. De Jong MJP, et al. Genetic Variants in SPINK1 , PRSS1, or CFTR Are Not Associated With The Development of Post ERCP Pancreatitis. Pancreas. 2025.
3. Sahin-Tóth M. Human cationic trypsinogen. Role of Asn-21 in zymogen activation and implications in hereditary pancreatitis. J Biol Chem. 2000.

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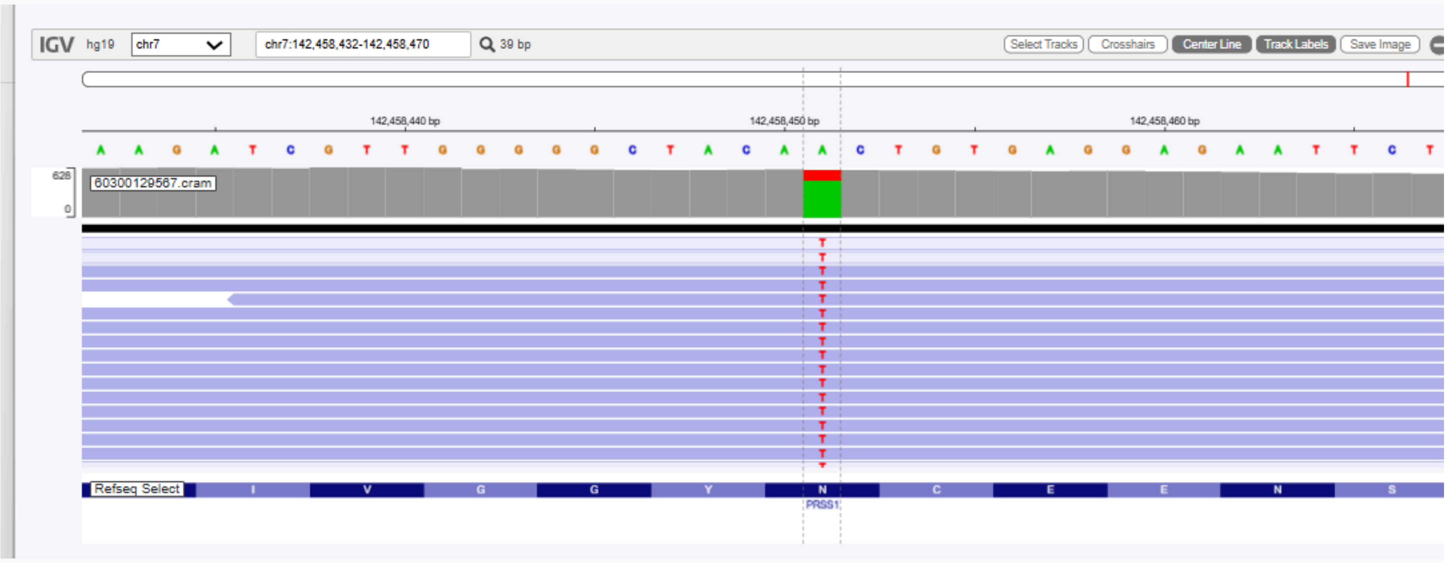
Variant Details

4.



Disease

PANCREATITIS, HEREDITARY [OMIM GENE ID: 276000]

PRSS1-related hereditary pancreatitis (HP) is characterized by episodes of acute pancreatitis (AP) and recurrent acute pancreatitis (RAP: >1 episode of AP), with frequent progression to chronic pancreatitis (CP). Mani from vague abdominal pain lasting one to three days to severe abdominal pain lasting days to weeks and requiring hospitalization.

NOTE

- **Variant classification is known to change with time and emerging evidence. Follow up with respect to changes in classification with your referring clinician/laboratory is recommended at 3-5 yearly intervals or as deemed required by your clinician.**
- **The clinical history section in the report contains concised details of the salient features provided for the patient. The analysis has been performed with all the clinical details provided and relevant HPO terms based on the same.**

Recommendations

1. **Clinical correlation as well as reverse phenotyping is recommended for all reports.**
2. **Genetic counseling for accurate interpretation of test results is recommended.**
3. **The reported findings are based on NGS analysis.**
4. **Analysis includes both single nucleotide (SNV) as well as copy number variant analysis (CNV).**
5. **Copy number variants when detected are included in the report.**
6. **Since CNV analysis is performed on a comparative basis, a negative result does not exclude the presence of a CNV.**
7. **The CNV pipeline is not validated for >3 exon copy number variants wherein detection is influenced by the underlying gene region and structure.**
8. **Variant calling (SNV and CNV) may be limited in low covered regions as well as in regions of low complexity and in pseudogenes.**
9. **Synonymous variants (not affecting splice site) as well as intronic variants are usually not reported.**
10. **Analysis and reporting is focussed on the provided phenotype and based on relevant HPO (Human Phenotype Ontology) terms as well as on genes associated with provided phenotype.**
11. **A genotype based analysis is also performed when the above yields negative results but reporting is limited to genes wherein current available evidence suggests a possible association with the provided phenotype.**
12. **It may not be possible to fully resolve certain details about variants, such as mosaicism, phasing, or mapping ambiguity.**
13. **Disease descriptions are included from OMIM, Genereviews and PUBMED indexed articles as and where applicable.**
14. **The test methodology currently does not detect large deletions/duplications, triplet repeat expansions and epigenetic changes. The test also does not include analysis of predictors for multifactorial, polygenic and/or complex diseases.**
15. **Phenotype variability may be due to modifying genetic/non-genetic factors and is not a part of the current analysis.**
16. **Candidate genes and genes with limited evidence are designated as genes of uncertain significance and all variants detected therein are classified as variants of uncertain significance.**
17. **Typically only variants at a depth of >/= 10X are reported. Lower depth variants may be false positives.**
18. **Detected variants in low complexity regions as well as variants at low depth and relatively low VAF should be reconfirmed by an alternate methodology.**
19. **CNV confirmation via MLPA or Exon array is recommended for copy number variants involving a single gene as well as for CNV <400kb. Only large constitutional CNV can be tested via other array platforms.**
20. **Variant depth/Total depth has been mentioned in summary of the variants.**
21. **Parental/Maternal testing as applicable is recommended for variants when detected for phasing (where applicable)**
22. **Segregation analysis (testing of multiple affected as well as unaffected members) of detected variants (if any) is recommended. Variant classification is subject to change after segregation.**
23. **ACMG secondary findings as well as carrier status of variants are only provided when requested.**
24. **Within carrier screening risk factors/alleles as well as hypomorphic variants may not be included**
25. **Interobserver as well as inter-laboratory variation is known with respect to variant classification due to the subjective nature of the provided criteria. Though the laboratory follows the updated recommendations provided by the ClinGen SVI as well as ACMG, independent assessment of variant classification by the referring clinician is recommended before decision making.**
26. **For prenatal samples analysis is limited to provided clinical phenotypes utilizing relevant HPO terms. Typically variants of uncertain significance are only introduced if the respective gene has been associated with the observed fetal phenotype. In solo fetal exomes, variants of uncertain deemed to be disease causing based on genotype characteristics may be included for further evaluation by the referring clinician. In prenatal scenarios trio fetal testing is strongly recommended to allow better interpretation of detected variants.**
27. **If the above results do not correlate completely with patient phenotype, additional testing is advised based on clinician's discretion.**
28. **Typically, heterozygous variants of uncertain significance in genes associated with autosomal recessive disorder are not reported in the proband. Such variants are included if relevant to phenotype in carrier screening.**

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Recommendations

29. A negative report does not exclude a genetic disorder due to inherent limitations of the assay design.
30. On the background of whole exome sequencing, analysis is limited to provided indications/ requested testing. Hence analysis is limited to a single gene if the same has been requested.
31. As a part of knowledge sharing initiative, all reported variants are submitted in de-identified form in the ClinVar database.
32. Extracted DNA if available after requisitioned testing will be stored as per recommendations. Please note that DNA may degrade over time and this may affect the quality of the stored sample.
33. Collected blood samples are not stored
34. Prenatal samples (AF/CVS/cord blood) are not stored. Extracted DNA if available is stored as acknowledged above.
35. As per PCPNDT fetal gender is not revealed.
36. Maternal cell contamination is recommended for prenatal and POC (product of conception) to ensure accuracy of test results. The same is performed only when requested and on the availability of the maternal sample.
37. Discrepant maternal cell contamination results may rise with use of donor gamete and hence information regarding the same should be provided to the laboratory.
38. Reproductive decision making is not recommended based on variants of uncertain significance.
39. Raw data can be transferred on request and after due consent/assent from the involved patient/ family. Additional charges will be applicable for the same.
40. The test performed by the laboratory with the assumption that the sample belongs to the person herewith mentioned in the requisition form and appropriate consent as well as prenatal counseling including test information has been provided by the referring clinician.
41. Repeat sampling may be required in case of gender discrepancy (unless the same can be attributed to an underlying scientific reason) as well as in rare cases where DNA/data quality prevents further analysis.
42. Reanalysis of data is recommended as deemed necessary by the referring clinician. Additional charges will be applicable for the same.

Technical Notes

Methodology - Massively Parallel Sequencing (Next Generation Sequencing): Genomic DNA from the submitted specimen was enriched for the complete coding regions and splice site junctions of genes listed below using a custom bait- capture system. Paired End Sequencing was performed with 2x100/2x150 chemistry. Reads were assembled and were aligned to reference sequences based on NCBI RefSeq transcripts and human genome build GRCh37/UCSC hg19. Data was filtered and analyzed to identify variants of interest and interpreted in the context of a single most damaging, clinically relevant transcript for the purpose of the report, indicated as a part of variant details. Enrichment and analysis focus on the coding sequence of the indicated transcripts, 5-10bp of flanking intronic sequence, and other specific genomic regions demonstrated to be causative of disease at the time of assay design. Deletion and duplication analysis is performed in cases when indicated but detected variations need to be confirmed by an alternate methodology. Sequence and copy number variants are reported according to the Human Genome Variation Society (HGVS).

Laboratory reporting protocol: The analysis is based on the provided phenotype: relevant HPO terms, curated gene panels and relevant literature is assessed for phenotype based analysis. Variant reporting is limited to exon regions and upto 10 basepairs within exon intron boundaries. Previously reported deep intronic and non coding variants will be included when detected at a depth more than 10X . Variant reporting is performed at a minimum depth of 10X. The gnomAD variant frequency reflects the liftover of hg38 to hg19.

For Mitochondrial Genome Sequencing (if requested): Only phenotype-related Pathogenic and Likely Pathogenic variations reported in the MitoMap database as well as literature are reported. Haplogroups are not analyzed. A list of variants other than the above are available on request. Analyzed genes include:MT-ND1, MT-ND2, MT-ND3, MT-ND4L, MT-ND4, MT-ND5, MT-ND6, MT-CYB, MT-CO1, MT-CO2, MT-CO3, MT-ATP6, MT-ATP8, MT-RNR2, MT-RNR1, MT-RNR2, MT-TA, MT-TR, MT-TN, MT-TD, MT-TC, MT-TE, MT-TQ, MT-TG, MT-TH, MT-TI, MT-TL1, MT-TL2, MT-TK, MT-TM, MT-TF, MT-TP, MT-TS1, MT-TS2, MT-TT, MT-TW, MT-TY, MT-TV.

Tools and Databases employed for analysis: Clinvar, OMIM, HGMD, UCSC genome browser, Uniprot, Ensembl, dbSNP, gnomAD, ExAC, Pubmed, Dgap, icgc, Kaviar, various bioinformatics analysis, predictive tools and disease specific databases used as available and appropriate. Such tools/databases would be mentioned wherever used.

REVEL: The REVEL score for an individual missense variant can range from 0 to 1, with higher scores reflecting greater likelihood that the variant is disease-causing.

Bioinformatics pipeline version: v16.1.1

Gene Coverage

Indication Based Analysis:

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
CASR	100%	CEL	96.2%	CFTR	100%	CLDN2	100%
CPA1	100%	CTRC	100%	GGT1	100%	PDX1	100%
PNLIP	100%	PRSS1	100%	PRSS58	100%	SBDS	100%
SPINK1	100%	UBR1	98%				

QC Metrics

Total aligned reads	99.95 %
Total reads	139.19 (M)
Total data generated	20.58 (Gb)
Total reads which passed mapping quality cutt-off	19.79 (Gb)

Reviewed By

UhKotecha

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